

Formal Equivalence Report — web_b5258d7e4ad0

Run: web_b5258d7e4ad0_optimize_20260914T144523Z

Per-Round Verification

Every candidate that reached formal checking, however the round ended:

Round 1: mac_unit (clock clk_a)

Candidate	Engine	Mode	Depth	Latency offset	Verdict	Log
1.1 (winner)	SymbiYosys	prove	20	+1	PASS	round_1/attempt_1/candidate_1/formal/equiv_sby_prove.log
1.2	SymbiYosys	prove	20	+1	PASS	round_1/attempt_1/candidate_2/formal/equiv_sby_prove.log
1.3	SymbiYosys	prove	20	+1	PASS	round_1/attempt_1/candidate_3/formal/equiv_sby_prove.log

Round verdict: ACCEPTED — Accepted [timing-closure]: net weighted improvement +5.0% (area -1.2%, timing +6.4%, power +0.0%).

Round 2: bm_dot4 (clock clk_c)

Candidate	Engine	Mode	Depth	Latency offset	Verdict	Log
1.1 (winner)	SymbiYosys	prove	20	+1	PASS	round_2/attempt_1/candidate_1/formal/equiv_sby_prove.log
1.2	SymbiYosys	prove	20	+1	PASS	round_2/attempt_1/candidate_2/formal/equiv_sby_prove.log
1.3	—	—	—	0	INFEASIBLE	—

Round verdict: ACCEPTED — Accepted [timing-closure]: net weighted improvement +37.7% (area -0.5%, timing +46.2%, power +7.7%).

Round 3: bm_fir6 (clock clk_b)

Candidate	Engine	Mode	Depth	Latency offset	Verdict	Log
1.1	SymbiYosys	bmc	24	+1	TIMEOUT	round_3/attempt_1/candidate_1/formal/equiv_sby_bmc.log

Candidate	Engine	Mode	Depth	Latency offset	Verdict	Log
2.1	SymbiYosys	bmc	64	+1	TIMEOUT	round_3/attempt_2/candidate_1/formal/equiv_sby_bmc.log

Round verdict: REJECTED — All 2 candidate(s) could not be verified in time (the equivalence checker timed out).

Chain of Proofs

PASS (chain of 2 proof(s): original $\rightarrow$ mac_unit $\rightarrow$ bm_dot4). Each round's module was proven equivalent against the previous round's design; equivalence of the final design to the ORIGINAL follows by transitivity, at the cumulative latency offsets reported below.

Cumulative Latency

Relative to the ORIGINAL design (each round's own proof is against that round's baseline, which is valid; these totals are what the surrounding logic must tolerate):

Module	Added latency
mac_unit	+1 cycle(s)
bm_dot4	+1 cycle(s)

Scope

Proven:

- Functional equivalence between each round's baseline RTL and its accepted candidate, at the module's port boundary, under the clocking/reset assumptions built into the equivalence miter (see each round's log for its exact depth/mode).
- Composed transitively: the final design is related to the ORIGINAL design by a chain of such proofs, one per accepted round, not by a single direct check.
- Pre-place-and-route static timing and PPA measurements (Yosys synthesis + OpenSTA), on the standard-cell library configured for this run.

Not proven / out of scope:

- **Clock-domain crossing (CDC) metastability.** The equivalence miter compares registered values at clock edges; it does not model metastable or glitchy behavior at an asynchronous crossing.
- **Reset behavior beyond the miter's own reset assumption.** The proof assumes reset is asserted long enough to bring both copies to a known state before comparison starts; reset sequencing bugs outside that window are not covered.
- **Post-place-and-route timing.** All timing numbers here come from a pre-P&R STA run; routing delay, clock skew and physical effects are not represented.